

ULTRON 2.7 Phase 12 Execution Report

Skills and local knowledge base

Project	ULTRON 2.7
Phase	Phase 12: Skills and Local Knowledge
Workspace	Local workspace checkout: U2.7
Status	Implemented and verified

Executive Summary

Phase 12 adds a reusable skill registry and local knowledge base to ULTRON 2.7. ULTRON can now list and run built-in skills, ingest safe local documents, search project knowledge, and expose skills, sources, and recent tasks through the web interface.

Skills and knowledge do not bypass safety. Skills validate their own schemas, pass through skill policy, and use the typed tool runtime for OS-facing actions. Knowledge retrieval provides context only; it never becomes execution authority.

Phase 12 Scope

- Add a skill registry with names, descriptions, schemas, risk levels, handlers, and examples.
- Add built-in notes, reminders, file search, project summary, and daily planning skills.
- Add a local knowledge base with safe-root enforcement, extraction, chunking, metadata, keyword search, and keyword-only embedding fallback.
- Reject obvious secrets during skill input validation and knowledge ingestion.
- Add skills and knowledge APIs.
- Add UI panels for skills, memory/knowledge, knowledge search, and recent task history.
- Add regression tests and architecture documentation.

What Was Implemented

Area	File(s)	Result
Skill registry	src/ultron27/skills.py	Schema validation, privacy checks, skill-level policy, built-in skills, examples, and skill-run audit records.
Knowledge base	src/ultron27/knowledge.py	Safe-root ingestion, Markdown/TXT/PDF/DOCX extraction where available, chunking, metadata, keyword search, and secret rejection.

Area	File(s)	Result
Runtime support	src/ultron27/runtime.py	Adds handle_tool_call() so skills can submit typed tool calls through validation, policy, executor, and audit.
Local API	src/ultron27/web_server.py	Adds skills and knowledge endpoints.
Interface	web/index.html, web/styles.css, web/app.js	Adds skills, memory/knowledge, local search, and recent-task panels.
Tests	tests/test_pipeline.py	Expands to 67 tests covering skills, knowledge, privacy, policy, and endpoints.

Safety Boundary

Boundary	Rule	Why It Matters
Skill schema	Inputs must match each skill schema.	Prevents ambiguous or unexpected skill payloads.
Skill policy	Medium and high-risk skills require confirmation.	Keeps reusable skills under the same permission model.
Tool runtime	OS-facing skills call handle_tool_call().	Preserves validation, policy, executor restrictions, and audit logging.
Knowledge paths	Ingested files must stay inside safe roots.	Prevents accidental indexing of private filesystem areas.
Knowledge content	Obvious secrets are rejected.	Prevents storing passwords, tokens, API keys, and credentials.
Retrieval	Search results are context only.	Knowledge cannot command ULTRON to execute tasks.

Verification Results

Check	Command	Result
Unit tests	\$env:PYTHONPATH='src'; python -m unittest discover -s tests	67 tests passed.
Compile check	python -m py_compile Phase 12 modules and scripts	Phase 12 Python modules compiled successfully.
Frontend syntax	bundled node --check web/app.js	Frontend JavaScript parsed successfully.
Dataset evaluation	python scripts/evaluate_dataset.py --split test	Intent 0.9965, tool 0.9965, argument exact match 0.9579, confirmation 1.0.
Safety regression	python scripts/evaluate_safety_regression.py	362 cases, tool accuracy 1.0, policy action accuracy 1.0.
API smoke	Local HTTP checks	Skills, knowledge ingest/search, and status panels returned expected payloads.

Recommended Next Step

- Add optional embeddings with a local provider and keep keyword search as fallback.
- Add source-linked answer generation that cites local chunks.
- Add a user-reviewed knowledge source manager for deleting and re-ingesting documents.